The Short, the Long, and the Consequences of VAR Analysis for Structural DSGE Estimation *

Lorenzo Del Col[ξ]

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Abstract

? advocate the use of long lags VARs for structural analysis, claiming that the truncation bias is a severe problem for the IRFs estimation. VARs can be used to create additional moments to estimate DSGE models minimizing the distance between their IRFs and therfore estimating the underlying structural parameters. In this study we ask what are the consequencies of enriching the lag structure of a VAR used for DSGE estimation. I replicate the results of **?** and I find that the both IRFs and Structural parameters are sensible to the chosen lag lenght. Matching the DSGE with a VAR(12) instead of the baseline VAR(4), I find that the Calvo Price parameters that differencieate the observationally equivalent **?** model between firm specific capital vs homogeneous capital are no longer statistically different from each other. The implied duration of the Calvo Price parameter is 2.5 times higher than the baseline, re opening the micro-macro discrepancy in price stickiness.

Keywords: Truncation Bias, Long Lag VARs, DSGE Estimation, IRFs, Minimum Distance Matching
JEL Codes: C32, C52, E32

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[ξ]University of Padova. Contact: lorenzo.delcol@studenti.unipd.it

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1 Introduction

1.1 Motivation

An important goal of monetary business cycle theoretical research is to explain the empirical evidence on the impact of economic shocks on macroeconomic variables. To study how monetary policy affects the economy, macroeconomics isolate changes in interest rate that are not systematic responses to economic conditions, but occur in a nonsystematic way. Since systematic variation in monetary policy is endogenous, understanding the causal effect of monetary policy on the economy requires considering such nonsystematic "monetary policy shocks" even if they are small or infrequent. **??** Dynamic Stochastic General Equilibrium (DSGE) models can be mapped in the form of State Space representation, which then can be retrived as Vector AutoRegressions (VAR) of infinite order. Impulse Response Function Matching Estimation (IRFME) is a limited informaiton approach that minimises the distance between VARs' IRFs and the one implied by a DSGE model. This technique requires that sample and theoretical IRFs are identified by restriction implied by the DSGE model, which requires a SVAR representation, so that the IRFs are informative fro the DSGE parameters **?**. If the reducced form VAR (RF-VAR) adequately captures the Data Generating Process (DGP), IRFs are optimal at each horizon. But if the RF-VAR is misspecified, the specification error naturally spill to strucutral statistics and will be compounded at each horizon, **??**. Since is not feasible to estimate a VAR(∞) as samples are finite, researchers truncate the VAR which becomes misspecified, and more mispecified the smaller the chosen lag lenght **??** "all VAR researchers should be concerned about lag truncation, regardless of the horizon of interest dictated by the research question. Truncation affects inference at all horizons, including the very short ones ($h \leq p$)" **?** "The overarching implication of our investigation is that, contrary to conventional wisdom, long-lag VARs are feasible in practice. Applied research should use them broadly because they safeguard against the significant problem that lag truncation implies for structural economic conclusions." **?**. "it is up to the researcher to show that her structural results are robust to significant increases in the lag length, far beyond the lag lengths typically considered in current practice." **?**

1.2 Contribution and Results

I analyse the consequences of enriching the lag structure of a VAR used to estiamte a DSGE with IRFs minimum distance estimation. I replciate **?** and find that the both IRFs and Structural parameters are sensible to the chosen lag lenght.

The Calvo Price parameters that differenciate the observationally equivalent **?** model between firm specific capital vs homogeneous capital are no longer statistically different

from each other. This goes against their main conclusion that firm specific capital model is the one that better fit the data.

(Maybe the Capital Adjustment Cost variable that since **?** researchers have started to use, is not fundamental to explain inflation inertia.)

1.3 Related Literature

I contribute in XXX breanches of research. The first is the one that study ... Long Lag VARs **????**
The second branch lies between the fast growing debate between Local Projections and VARs IRFs (VARs and LPs) **???**
The third, State Space rappresentation of DSGE **?????**
The fourth, fundamentalness and non-fundamentalness, information sufficiency and deficiency **?** (LIPPI RIECHILIN 1993 ECC)

1.4 Outline

The rest of the work is organised as follows. In Section 2 I present the Truncation Bias problem, the theory that maps DSGE into VAR(inf) that suggests to increase the lag length. In section 3 I summaryse briefly the **?** DSGE. In section 4 I present the empirical results and in section 5 I conclude.

2 Truncation Bias: Illustration of the Problem

VARs has at least three source of bias: Small Sample Bias (cit uno degli anni 50, l'ho visto penso su Chari et al 2008, anche loro li posso citare qua), Identification Bias **?**, and Truncation Bias **?????**, which is the principal focus of this work.

To explain it, I follow the single equation and then generalization to a VAR example used by **?**.
Let's consider a DGP

$$y_t = x_{1t}\beta_1 + x_{2t}\beta_2 + \epsilon_t \tag{1}$$

where the variable y_t is determined by two exogenous variables, x_{1t} and x_{2t} and a shock ϵ_t. If y_t is estimated with a regression omitting one of the two exogenous variables, say x_{2t},

$$y_t = x_{1t}b_1 + e_t \tag{2}$$

the omission leads to biaesd point estimate (unless $x_{1t} \perp x_{2t}$).

$$\mathbb{E}(b_1) \neq \beta_1 \tag{3}$$

with an always upwardly biased variance estimate.

$$s^2 > \sigma^2 \tag{4}$$

Translating this to VARs

$$Y_t = B_1 Y_{t-1} + ... + B_p Y_{t-p} + U_t, \mathbb{E}[U_t U_t'] = \Sigma_U \tag{5}$$

$$B(L) = B_1 L + ... + B_p L^p \tag{6}$$

which has $p << p^*$, where p^* denotes the true unknown lag length, the VAR will suffer from truncation bias. Lag truncation cause bias in reduced form coefficients $B(L)$ and in the reduced form variance covariance matrix Σ_U. Which are both ingredients of IRFs

$$IRF = f(B(L); \Sigma_U) \tag{7}$$

If the components are biased by lag truncation, so will the IRFs. **?** explores the trivial solution to truncation bias, that is the dramatic increase of lag length. They find evidence against the conventional wisdom that uncertainty would become pervasive as the lag choice would increase the number of parameters rapidly, hence reducing the degrees of freedom and making confidence interval width prohibitively large. But the standard intuition is only part of the story, because increasing the lag choice not only reduces the misspecification in the well known bias reduction, but also it reduces the variance. This reduction in variance works against the imprecision resulting from increased parametrization **?**.

3 DSGE State Space Representation and VARMA

Simple theoretical derivation describing that even if the **?** is stochastic singular (and defend the fact that most of the SVAR analysis is done with stochastic singular models (more observables than shocks), **?**) and explain that the solution of the DSGE in State Space can be represented as a VARMA **?**, hence the use of VAR with long lags is justified even if the ABC and D **?** representation of the VAR(∞) does not exist.

4 Theoretical Model

? noticed that macro modelers to account for inertia in inflation, were used to embed implausible assumption on a priori grounds or even directly in conclift with micro data. For example, there are many New-Keynesian models in which non optimizer firms index their prices in lagged inflation that are able to account for inflation inertia. But assuming that firms re optimise their price every six quarters or even less. In other versions of

NK models in which non optimisers firms are not allowed to index to past inflation, the implied duration is even greater i.e. firms change prices every two years.
This long implied duration of prices for the non optimiser firms is in contrast with the micro literature which argue that firms change prices more frequently than once every two quarters. (THEY CITE: This property contrasts sharply with findings in Bils and Klenow (2004), Golosov and Lucas (2007) and Klenow and Kryvtsov (2008). some more recent literature can be cited here.).
? create a model, with non optimiser firms that index their prices to lagged inflation, which is consistent with the evidence that inflation is inertial, even though firms re optimise prices on average every 1.5 quarters. They are able to model aggregate inflation even if firms re optimise prices frequently. The inertia is explained by the fact that, when firms do change prices, they do it by a small amount because each firm's short run marginal cost curve is increasing in its own output. And this positive dependency reflects on the assumption that in any given period, a firm's capital stock is predetermined.
In their model firm's capital stock is pre-determined, and all the factors of production, including capital, cannot be instantly transferred across firms. The assumption behind is that capital is completely firm-specific, and can be changed over time by varying the rate of investment. This feature was not standard in business cycle models.

? conduct the analysis using two versions of the **?** DSGE in which, in one model the capital is *homogenoeous* as in standard models, whereas in the other is *firm-specific*. The CEE model is a special case of ACEL when only monetary shocks are considererd **?**. In the log-linearized equilibrium equations the two models are equivalent except for the equation that relates inflation to the firms' marginal costs. The form of this equation is identical in both the models: the change in inflaiton at time t is equal to discounted expected change in inflation at time $t+1$ plus a reducef form coefficient γ, being the Phillips Curve slope, multiplying the time t economy-wide average marginal cost. The two models are observationally equivalent, and the difference lies in the mapping between the structural parameters inside the Phillips Curve.

$$\Delta\hat{\pi}_t = \mathbb{E}[\beta\Delta\hat{\pi}_{t+1} + \gamma\hat{s}_t|\Omega_t] \tag{8}$$

where

$$\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\chi \tag{9}$$

and Δ is the first difference operator. The information set Ω_t includes the current realization of the technology shocks, but not the current realization of the innovations to monetary policy. The variable s_t denotes the economy-wide average marginal cost of

production, in units of the final good. It can be re-written as

$$\Delta\hat{\pi}_t = \mathbb{E}[\beta\Delta\hat{\pi}_{t+1} + \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\chi|\Omega_t] \tag{10}$$

From (ACEL2004 VEDERE CHE PAER è E CITARLO). in the *homogeneous* capital model, $\chi = 1$, whereas, in the *firm-specific* capital model χ is a particular non-linear function of the parameters of the model. The two models are parametrised in terms of γ rather than ξ_p. This guartees that the list of structural parameters remains identical and the two models are observationally equivalent with respect to aggregate prices and quantities. This means that there is no need to take a stand on which model version we are working with at the estimation stage of our analysis **?**.

To choose the model parameter values of the the models they use limited information estimation technique IRFME following **?** citare ROTOBERG AND WOODFORD 1997 E QUALCHE ALTRO, miminise the differences between the dynamic response to shocks in the model and the analog objects estimated using a ten variables VAR of post-war U.S. time series. IRFME is used not only when there is stochastic singularity, but also when the modeler want to take a positon on the identification of a subset of shocks, not every shock in the system. (CIT EFREM IN UNA MAIL, trovare una citazione di un paper) The VAR identification assumptions are such that satisfy the one from the economic model: only tech shocks can affect productivity in the long run; capital embodied technology shock is the only one that can affect price of investment goods in the long run, and monetary policy shock can have contemporaneous effects only on the interest rate as in standard zero short run restrictions.

MY DISCUSSION The quality of the estimation strategy depends on the ability of identified VARs to generate reliable estimates of the dynamic response of economic variables to shocks [footnote 10, pag 3.]ACEL2011. **?**[pag. 83] "if the VAR adequately captures the DGP, IRFs are optimal at all horizons. If the VAR is misspecificed, however, then the specification errors will ve computed at each horizon" **?** says that is responsibility of the short-lag VAR users to demonstrate if their results are robust to long-lag VARs.

5 Empirical Results

Descrizione dei risultati che trovo dai miei esercizi long lag VARs. Immagini IRFs, DSGE Parameters table and implied time duration.

Generalized variance and so on and so forth.

5.1 Implied Duration from Calvo Price Table: lags 4 8 12

Table 1: Implied Average Time between Re-Optimization

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model			
Calvo parameter ($\xi_{p,hom}$)	0.82	0.82	0.90
Implied duration (quarters)	**5.6**	**5.6**	**9.6**
Firm-Specific Capital Model			
Calvo parameter ($\xi_{p,fs}$)	0.34	0.64	0.89
Implied duration (quarters)	**1.5**	**2.8**	**9.0**

- **Baseline VAR ($p = 4$)**: Homogeneous capital implies price changes every **5.6 quarters**.
- **Firm-specific capital**: Drastically reduces this to **1.5 quarters**, matching microeconomic evidence.

Note: Implied duration is calculated as $D = 1/(1 - \xi_p)$.

6 Conclusion

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A Replication Files and Code

Detail where the reader can find your code (e.g., a GitHub link).

B Full Description of ACEL Model

C MonteCalvo

MonteCarlo simulation for the SE of the Implied Calvo Parameters, derived by a simulation of a joint normal because in the IRFME they use the Deltha Method which assumse joint normality

D Additional Figures

D.1 ACEL 2005 VAR Variables

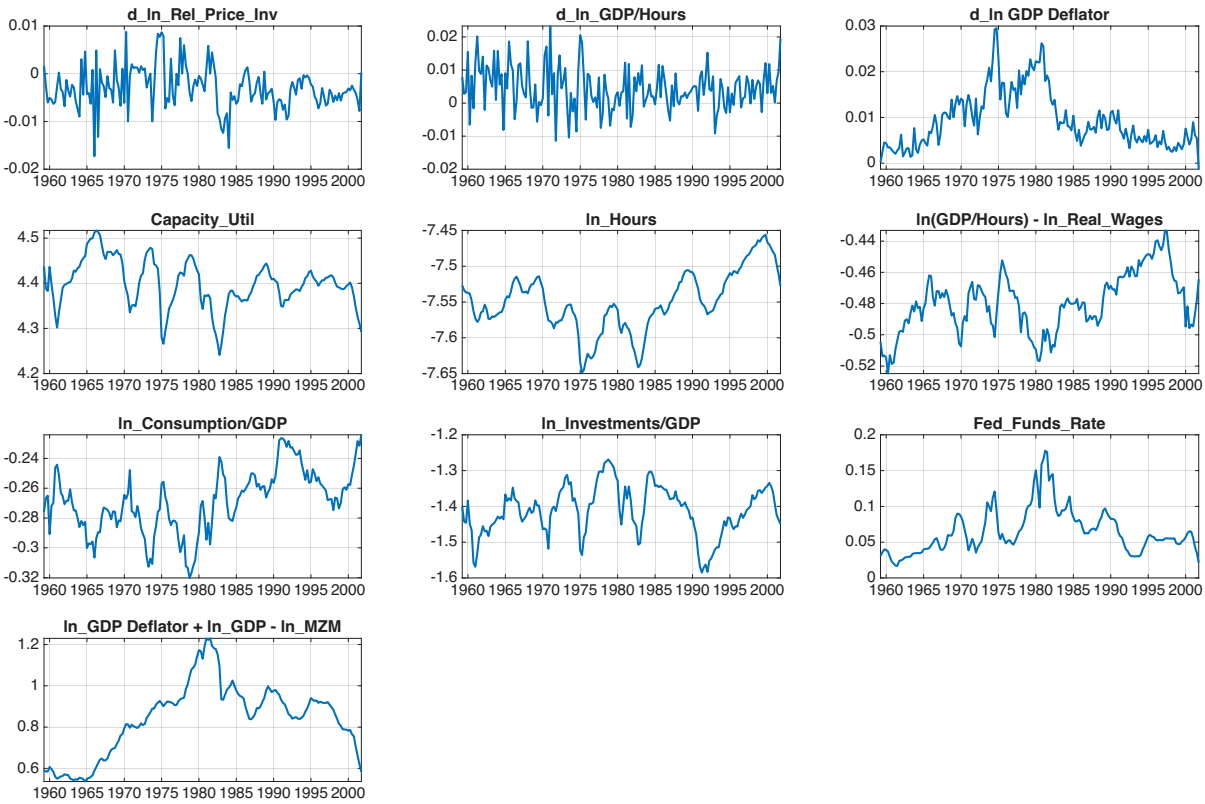


Figure 1: Raw Historical Series